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# **Bank Loan Portfolio Performance Report**

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## **1. Introduction**

*This report presents a structured analysis of the bank's loan portfolio using a comprehensive **Bank Loan Reporting** framework. The objective is to monitor lending activity, assess portfolio health, and support data-driven decision-making through clearly defined performance indicators and interactive dashboards.*

*The analysis is built on a bank loan dataset containing loan-level information such as application identifiers, funding amounts, repayments, interest rates, borrower characteristics, and loan status. Using SQL-based data preparation and interactive dashboards, the report delivers both high-level portfolio views and granular drill-downs.*

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## **2. Objectives**

*The key objectives of this project are to:*

- *Track **overall lending activity** through total loan applications and funded amounts.*
- *Monitor **cash inflows and repayment performance** via total amount received.*
- *Understand the **pricing and risk profile** of the portfolio through average interest rate and average debt-to-income ratio (DTI).*

- Classify loans into **Good Loans** and **Bad Loans** based on loan status to evaluate portfolio quality.
  - Analyse trends across **time, geography, loan characteristics, and borrower attributes**.
  - Provide a **details view** that consolidates all critical loan data for operational and analytical use.
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### **3. Data and Methodology**

#### **3.1 Data**

*The underlying dataset includes, among others, the following fields:*

- *Loan application ID*
- *Issue date of the loan*
- *Funded amount (principal)*
- *Total amount received (repayments)*
- *Interest rate*
- *Debt-to-income ratio (DTI)*
- *Loan status (e.g., Fully Paid, Current, Charged Off)*

- *Borrower's state*
- *Loan term (e.g., 36 months, 60 months)*
- *Employment length*
- *Loan purpose (e.g., debt consolidation, credit card refinancing)*
- *Home ownership status (e.g., own, rent, mortgage)*

### **3.2 Tools and Approach**

- **SQL** is used to aggregate and filter the data, compute Month-to-Date (MTD) and Month-over-Month (MoM) metrics, and generate segment-wise summaries (by status, state, term, purpose, etc.).
- **Power BI** is used to build the data model and create three dashboards:
  - **Dashboard 1: KPI & Quality Overview** (portfolio-level KPIs and Good vs Bad loan metrics)
  - **Dashboard 2: Overview** (visual analysis by time, geography, and borrower/loan characteristics)
  - **Dashboard 3: Details** (loan-level drill-down)

*All measures and visualizations are built on top of a consistent SQL logic layer.*

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## **4. KPI Framework (Dashboard 1 – Portfolio Performance)**

*Dashboard 1 focuses on high-level performance indicators, including MTD and MoM dynamics. The reporting period is defined using the issue date of the loans, with current MTD and prior month values compared to monitor trends.*

### **4.1 Total Loan Applications**

- **Definition:** Total number of loan applications recorded in the dataset, based on distinct loan IDs.
  
- **Breakdowns:**
  - Overall total
  
  - **Month-to-Date (MTD)** total loan applications
  
  - **Previous Month-to-Date (PMTD)** loan applications
  
  - **MoM change (%)** comparing MTD with PMTD

#### **Interpretation:**

*This metric shows how demand for loans is evolving. Rising MoM applications may indicate increased demand or more aggressive marketing, while a decline could signal tightening underwriting, reduced demand, or seasonality.*

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### **4.2 Total Funded Amount**

- **Definition:** Total principal amount disbursed as loans.

- **Breakdowns:**

- Overall total funded amount
- MTD funded amount
- PMTD funded amount
- MoM change (%) in funded amount

**Interpretation:**

This metric reflects the bank's **credit deployment**. Comparing funded amounts to application counts also helps distinguish between **demand growth** and **approval / underwriting changes**. A faster increase in funded amount compared to applications may indicate larger average loan sizes or relaxed credit limits.

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#### **4.3 Total Amount Received**

- **Definition:** Total amount repaid by borrowers (all payments received).

- **Breakdowns:**

- Overall amount received
- MTD amount received

- PMTD amount received
- MoM change (%) in collections

**Interpretation:**

*This is a key **cash flow and collections** indicator. Healthy portfolios generally show a stable or growing level of repayments, aligned with the growth of funded amounts. Divergence between rapid funding growth and slower repayments can be an early warning signal of **emerging credit risk**.*

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#### **4.4 Average Interest Rate**

- **Definition:** Average interest rate across all loans, expressed as a percentage.
- **Breakdowns:**
  - Overall average interest rate
  - MTD average interest rate
  - PMTD average interest rate
  - MoM change (absolute or percentage)

**Interpretation:**

*Average interest rate reflects the **pricing of the loan book**. Increases may indicate shifts towards riskier borrowers or higher-yield products, while decreases may suggest a focus on*

lower-risk segments or competitive pricing. Tracking MoM trends helps management understand whether new originations are changing the risk-return profile.

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#### **4.5 Average Debt-to-Income Ratio (DTI)**

- **Definition:** Average DTI of borrowers, typically expressed as a percentage (borrower monthly debt obligations relative to income).
  
- **Breakdowns:**
  - Overall average DTI
  
  - MTD average DTI
  
  - PMTD average DTI
  
  - MoM change in DTI

#### **Interpretation:**

DTI is a proxy for **borrower affordability and leverage**. A rising average DTI over time may indicate that the bank is gradually approving riskier borrowers, which can translate into higher future default risk. A stable or declining DTI suggests tighter credit standards or a shift towards more financially resilient customers.

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### **5. Portfolio Quality: Good vs Bad Loans**

To assess portfolio quality, loans are classified based on their status:

- **Good Loans:** Loans with a status of “Fully Paid” or “Current”
- **Bad Loans:** Loans with a status of “Charged Off”

*This classification is used to compute Good vs Bad Loan key performance indicators.*

## **5.1 Good Loan KPIs**

### **1. Good Loan Application Percentage**

- *Share of total loan applications that are classified as Good Loans.*
- *Indicates the **quality mix** of the portfolio by volume.*

### **2. Good Loan Applications**

- *Total number of applications that fall under Good Loans.*
- *Shows how many loans are either fully repaid or currently performing.*

### **3. Good Loan Funded Amount**

- *Total principal funded for Good Loans.*
- *Represents the share of capital deployed into healthy, performing assets.*

### **4. Good Loan Total Received Amount**

- *Total repayments collected from Good Loans.*



- Reflects the **cash generation** from the performing portion of the portfolio.

**Insight:**

A high Good Loan percentage and strong Good Loan funded and received amounts indicate a **robust, well-performing loan book**. This is the core revenue-generating segment of the portfolio.

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## **5.2 Bad Loan KPIs**

### **1. Bad Loan Application Percentage**

- Percentage of total applications classified as Bad Loans (Charged Off).
- Directly linked to **default incidence** in the portfolio.

### **2. Bad Loan Applications**

- Total count of applications that have resulted in charge-offs.

### **3. Bad Loan Funded Amount**

- Total principal associated with Charged Off loans.
- Represents **capital at loss or high risk**, useful for loss provisioning and risk capital planning.

### **4. Bad Loan Total Received Amount**

- *Total repayments received from Charged Off loans (partial recoveries).*
- *Helps evaluate **recovery rates** and collections effectiveness on bad accounts.*

***Insight:***

*By monitoring the ratio of Bad to Good loans and the funded amounts associated with each, the bank can track **credit risk, portfolio deterioration, and recovery performance** over time.*

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## **6. Loan Status Grid View**

*A grid view is created to provide a **status-wise snapshot** of the portfolio. For each loan status (e.g., Fully Paid, Current, Charged Off), the following indicators are summarised:*

- *Total loan applications*
- *Total funded amount*
- *Total amount received*
- *MTD funded amount*
- *MTD amount received*
- *Average interest rate*
- *Average DTI*

### **Usage and Insights:**

- Compare **performing statuses** (Fully Paid, Current) against **non-performing** (Charged Off).
- Understand whether **Charged Off loans** tend to have higher average interest rates or higher DTI, indicating pricing and underwriting patterns associated with risk.
- Evaluate the status-wise contribution to both **portfolio size (funded amount)** and **cash flow (amount received)**.

*This grid helps risk and business teams quickly identify where risk is concentrated and how each status contributes to the overall performance.*

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## **7. Overview Dashboard (Dashboard 2 – Visual Analytics)**

*Dashboard 2 provides multi-dimensional, visual analysis of the portfolio using different chart types and segmentations.*

### **7.1 Monthly Trends by Issue Date (Line Chart)**

- **Metrics:**
  - *Total loan applications*
  - *Total funded amount*
  - *Total amount received*

- **X-axis:** Month (derived from loan issue dates)
- **Y-axis:** Metric values

**Insights:**

- Identifies **seasonality** in lending and repayments.
  - Highlights months of **peak originations** versus months of subdued activity.
  - Allows comparison between the pace of **funding growth** and **collection growth** over time.
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## **7.2 Regional Analysis by State (Filled Map)**

- **Metrics:** Total loan applications, total funded amount, total amount received
- **Dimension:** State

**Insights:**

- Identifies **high-volume lending regions** and under-penetrated states.
- Allows comparison of **funded amount vs collections** by region, which is useful for spotting **localised risk clusters** or collection issues.

- *Supports geographically targeted strategies in terms of marketing, underwriting, and collections.*
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### **7.3 Loan Term Analysis (Donut Chart)**

- **Metrics:** *Total loan applications, total funded amount, total amount received*
- **Segments:** *Loan terms (e.g., 36 months, 60 months)*

#### **Insights:**

- Shows the **distribution of portfolio volume and value** across different tenures.
  - Allows assessment of whether longer-term loans represent a disproportionate share of **funded exposure** or **repayment risk**.
  - Helps in understanding the bank's **tenor strategy** and its implications on liquidity and risk.
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### **7.4 Employee Length Analysis (Bar Chart)**

- **Metrics:** *Total loan applications, total funded amount, total amount received*
- **Axis:** *Employment length categories (e.g., <1 year, 1–5 years, 10+ years)*

### **Insights:**

- Evaluates how **employment stability** correlates with demand and performance.
- Helps answer questions such as:
  - Do borrowers with longer employment histories receive **larger loans**?
  - Are charge-offs more frequent in borrowers with **shorter employment tenure**?

This analysis can guide **underwriting criteria** related to employment history.

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### **7.5 Loan Purpose Breakdown (Bar Chart)**

- **Metrics:** Total loan applications, total funded amount, total amount received
- **Axis:** Loan purpose categories (e.g., debt consolidation, credit card refinancing, home improvement)

### **Insights:**

- Clarifies **why customers are borrowing**, by volume and value.
- Detects which loan purposes drive **most of the funded amount** and whether certain purposes are associated with **weaker repayment performance**.
- Supports product strategy (e.g., whether to promote or tighten certain categories).

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## 7.6 Home Ownership Analysis (Tree Map)

- **Metrics:** Total loan applications, total funded amount, total amount received
- **Hierarchy:** Home ownership categories (own, rent, mortgage)

### **Insights:**

- Shows how **ownership status** relates to lending and repayment behaviour.
- Enables comparison of **loan sizes** and **collection performance** across owners, renters, and mortgagors.
- Helps in aligning lending policies and marketing with target home ownership segments.

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## 8. Details Dashboard (Dashboard 3 – Loan-Level View)

The Details Dashboard consolidates all key attributes for each loan, providing a **single, comprehensive view** of:

- Loan ID and issue date
- State, term, purpose, and home ownership
- Loan amount and total amount received

- *Interest rate and DTI*
- *Loan status and Good/Bad loan classification*

*Interactive filters allow users to slice by:*

- *Date range*
- *Loan status (including Good vs Bad loan segments)*
- *State*
- *Term*
- *Purpose*
- *Home ownership*
- *Employment length*

***Use Cases:***

- *Detailed review of specific **segments of concern** (e.g., high DTI, high interest, charged-off loans).*
- *Operational support for **collections and risk teams**.*



- Data extraction for **further modelling and analysis** (e.g., probability of default models, cohort analysis).
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## 9. Key Business Insights and Value

Even without showing specific numeric values, the structure of this reporting system provides substantial business value:

1. **Standardised KPI Definitions**

– Clear, consistently calculated metrics for applications, funded amounts, collections, interest rate, and DTI create a single source of truth for performance monitoring.

2. **Transparent View of Portfolio Quality**

– Good vs Bad loan segmentation and status-wise metrics allow management to measure **portfolio health, loss exposure, and recovery performance** in a structured way.

3. **Rich Multi-Dimensional Analysis**

– Time, region, borrower profile, loan term, purpose, and home ownership are all available dimensions, enabling a deep understanding of **where growth and risk are coming from**.

4. **Actionable Dashboards**

– High-level KPIs, visual overviews, and drill-down capabilities turn raw data into **operationally useful insights**, informing strategy across product, underwriting, collections, and regional focus.

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## **10. Conclusion**

*The Bank Loan Report project delivers a complete analytics framework for monitoring and managing the loan portfolio. By combining well-defined KPIs, Good vs Bad loan classification, status-wise grids, multi-dimensional visual analyses, and a detailed loan-level dashboard, the system supports:*

- *Continuous tracking of **portfolio growth and quality***
- *Early detection of **emerging risks***
- *More informed, data-driven decisions in **lending strategy, pricing, and risk management***

*This framework can be further extended by integrating additional variables such as credit scores, behavioural indicators, and external macroeconomic data, as well as by developing predictive models for defaults, loss given default, and customer lifetime value.*